Syllabus

01204465 Introduction to Data Mining, Data Analytics & Knowledge Discovery

Semester 1/2569

Department: Computer Engineering, Faculty of Engineering

Instructor: Assoc. Prof. Kitsana Waiyamai, Ph. D.

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Teaching Assistant: Jetana Somkamnueng (P'Jame)

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Google Classroom Code: aulxievt

Class Schedule: Mondays 09:00–12:00

Location: Room s601, 6st Floor, Building E11

Software Used: RapidMiner & Python (Jupyter Notebooks)

Course Description

This course introduces core concepts in data mining, data analytics, and knowledge discovery. It emphasizes the complete process of extracting actionable knowledge from data, starting from data exploration and preprocessing to applying analytics and mining techniques. Topics include pattern discovery, classification, clustering, and advanced analytics on complex data. Both theoretical foundations and hands-on practice using industry tools are included.

Learning Outcomes

By the end of the course, students will be able to:

- Understand and apply the key stages of the knowledge discovery and data analytics process.
- Perform data cleaning, transformation, exploration, and visualization.
- Apply data mining and analytics techniques such as association rule mining, classification, and clustering.
- Interpret patterns and insights from structured and complex data.
- Use data analytics tools such as RapidMiner and Python effectively.
- Communicate findings and insights derived from data clearly and accurately.

Textbooks & References

- Shmueli et al., Data Mining for Business Intelligence: Concepts, Techniques, and Applications in Data Analytics, Wiley.
- Han, Kamber & Pei, Data Mining: Concepts and Techniques, 3rd ed., Morgan Kaufmann.
- Udit Agarwal, Data Mining and Data Warehousing, S.K. Katarina & Sons, 2016.

Course Outline and Weekly Schedule

Week	Topic	Speaker
1	Introduction to Data Mining, Data Analytics, and KDD	Kitsana
2	Data Preprocessing Techniques for Analytics	Kitsana
3	Exploratory Data Analytics (EDA)	Kitsana
4	Data Warehousing Design for Analytics	Kitsana
5	Data Warehousing Implementation	Kitsana
6	Pattern Discovery: Association Rule Mining	Kitsana
7	Classification Techniques (e.g., Decision Trees, k-NN)	Kitsana
8	Midterm Exam (Lecture + Lab, On-site)	—
9	Group Presentations (Analytics & Preprocessing Pipeline)	Students
10	Clustering Techniques (e.g., k-means, DBSCAN)	Kitsana
11	Evaluation & Validation of Analytics Models	Kitsana
12	Complex Data Mining & Advanced Data Analytics - I	Kitsana
13	Complex Data Mining & Advanced Data Analytics - II	Invited Speaker
14	Final Group Project Presentations	Students
15	Final Exam (Lecture + Lab, On-site)	—

Evaluation Policy

Component	Weight
Group Assignment (Project Proposal + Presentation)	20%
Midterm Exam	30%
Final Exam	30%
Homework Assignments	10%
Quizzes & Class Participation	10%